# Paleomagnetism of Welded Tuffs of the Yellowstone Group

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Two of the three ash flow tuffs of the Yellowstone Group are stably magnetized throughout their extent: the Lava Creek tuff (0.60 m.y.) in a normal direction and the Mesa Falls tuff (1.22 m.y.) in a reversed direction. In contrast, much of the Huckleberry Ridge tuff, which was erupted during a polarity transition or field excursion 1.92 m.y. ago and initially magnetized in an intermediate direction ($D$ = 214.5°, $I$ = −2.2°, and $\alpha_{95}$ = 4.1°), has been partially remagnetized in a normal field. Much of the overprint of the Huckleberry Ridge tuff is carried in low-temperature oxidation products with low coercivities and low blocking temperatures, so that it is commonly removed in peak alternating fields less than 200 Oe or at temperatures less than 400°C. Hematite, having higher coercivities and blocking temperatures, is partly responsible for remagnetization in some specimens, but neither alternating field nor thermal demagnetization enables recovery of the original thermal remanent magnetization in these samples. The chemical overprinting is most pronounced in the devitrified tuff of the flow interior. Stable magnetization is observed in the Huckleberry Ridge tuff in the less porous densely welded portions at the base and boundaries of partial cooling units and in the tuff of the distal margins, where groundwater influences were minimal. Paleomagnetic directions of chilled margins within compound cooling units of the Lava Creek and Huckleberry Ridge tuffs support geological evidence that eruption of each ash flow sheet was rapid and that the source calderas were formed during a short period of time.

## INTRODUCTION

Although rhyolitic welded ash flow sheets (welded tuffs) occur worldwide, are voluminous, and range in age from Precambrian to Holocene, they are magnetically among the least studied of the extrusive rocks. Most magnetic studies of Quaternary and Tertiary welded tuffs have stressed that these rocks are well suited for paleomagnetic research because of their generally high magnetic stability [*Doell et al.*, 1968; *Dalrymple et al.*, 1965; *Grommé et al.*, 1972]. Nevertheless, magnetic stability is not universal among all tuffs and is not well understood with respect to the mineralogy of the magnetic phases. With the exception of a few investigations [e.g., *Hatherton*, 1954*a*, *b*; *Gose*, 1970; *Nairn et al.*, 1975], little work has been done on the rock magnetic properties of welded tuffs, in contrast to extensive research on the rock magnetism of basaltic rocks [e.g., *Watkins and Haggerty*, 1967; *Wilson et al.*, 1968; *Ade-Hall et al.*, 1964, 1968*a*, *b*; *Larson et al.*, 1969].

In an attempt to understand the magnetic properties of rhyolitic ash flow sheets better a paleomagnetic study of the three welded tuffs of the Yellowstone Group was undertaken. This paper summarizes their rock magnetic characteristics, the nature of the magnetic phases, and the effects of high- and low-temperature oxidation on magnetic properties. Because one of the three Yellowstone tuffs was partly remagnetized long after its initial cooling, the relationships between authigenic magnetic phases, stability of remanence and remanent directions, and certain physical properties of the ash flow sheet are discussed in detail. Therefore this paper bears upon the problem of chemical remagnetization, recognized in igneous and sedimentary rocks of Tertiary age and older [*Roy and Park*, 1972; *Larson and Walker*, 1975; *Watkins*, 1967] but generally undocumented for younger extrusive rocks.

In addition, paleomagnetic directions of tuff from an extensive areal and vertical sampling are given in order to document the amount of paleomagnetic dispersion to be expected in young voluminous and little disturbed ash flow tuffs. Paleomagnetic determinations have also been helpful in distinguishing between the three tuffs of the Yellowstone Group and therefore are useful in deciphering the late Cenozoic volcanic stratigraphy of the Yellowstone rhyolite plateau and eastern Snake River Plain.

Finally, information about the time interval between eruptions of cooling units of tuff can be obtained by comparing the mean directions of the separate cooling units within a section. If the time interval of eruption of a thick section of tuff were small in relation to paleomagnetic secular variation (PSV), small or negligible differences in remanence directions would be seen throughout the unit; if, however, eruption occurred over a long period of time in relation to PSV, significant differences in the remanence directions would be expected.

## GEOLOGIC SETTING

Extensive Quaternary eruptives in the Yellowstone National Park and adjacent Island Park area of Wyoming, Idaho, and Montana mark the present easternmost extent of volcanism which has been migrating eastward along the Snake River Plain (Figure 1) (R. L. Christiansen, personal communication, 1968; *Armstrong* [1974]). The sequence of volcanics in the Yellowstone region consists of three distinct volcanic cycles, each primarily represented by a major ash flow sheet. These are the Huckleberry Ridge tuff (1.92 m.y.), the Mesa Falls tuff (1.22 m.y.), and the Lava Creek tuff (0.60 m.y.), which together comprise the Yellowstone Group [*Christiansen and Blank*, 1972]. The eruption of each volcanic sheet was associated with the collapse of a caldera of enormous proportions [*Christiansen and Blank*, 1972]. The three tuffs of the Yellowstone Group and rocks of related volcanic cycles are fully described elsewhere by *Christiansen and Blank* [1972], but characteristics pertinent to this paleomagnetic study are briefly summarized in the following section.

Within complete or nearly complete exposures the tuffs can be identified by the presence of partial cooling boundaries, which are reflected by significant differences in the degree of welding and by variations in phenocryst content. Where thick exposures are absent, local distinguishing characteristics may be used for identification of the tuffs.

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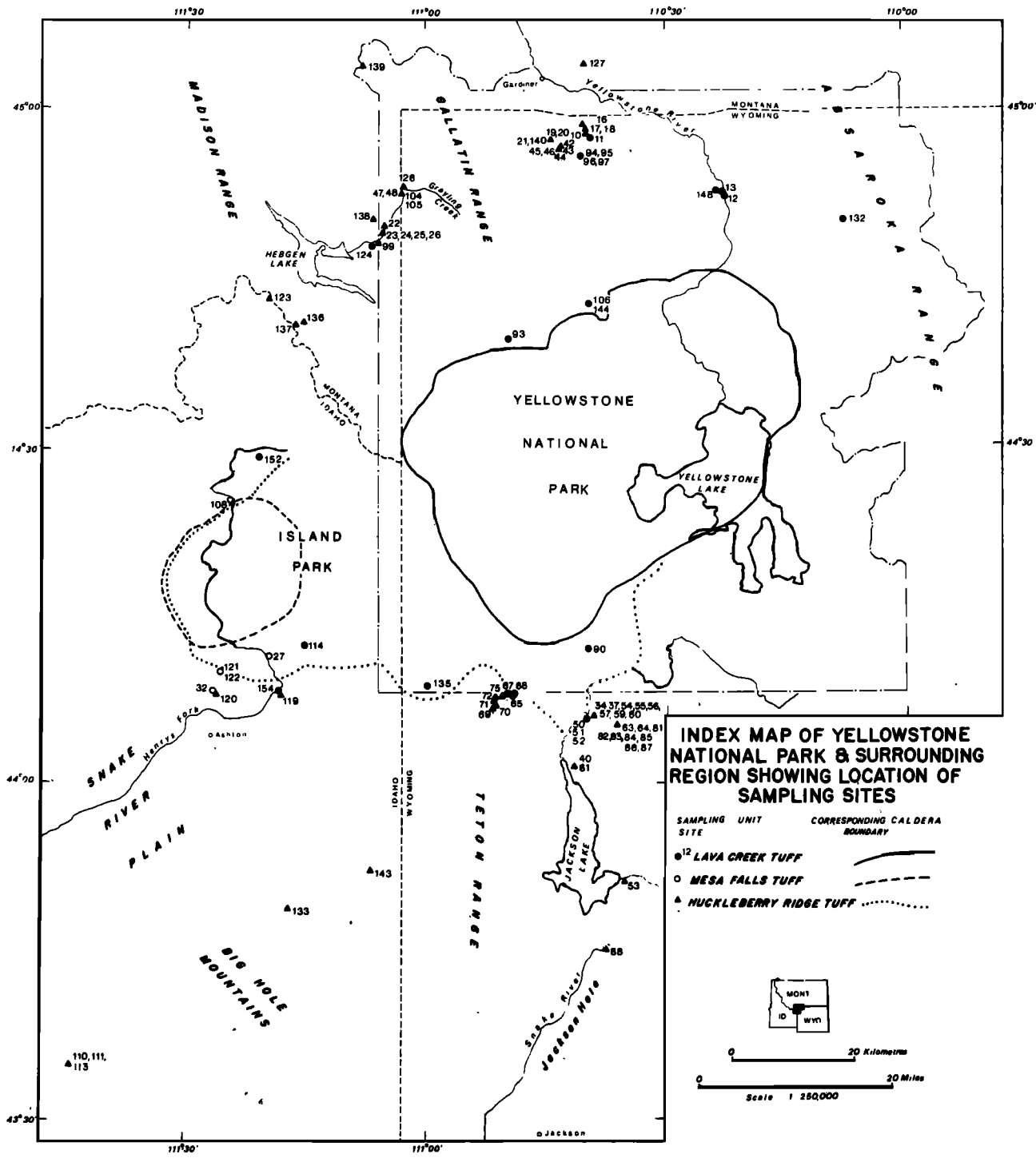



Fig. 1. Location map of Yellowstone National Park and surrounding area showing outlines of the major calderas referred to in the text (solid line, Yellowstone caldera; dashed line, Henrys Fork caldera; and dotted line, caldera segments of the Huckleberry Ridge tuff) and collecting localities of the three tuffs of the Yellowstone Group listed in the text.

The Huckleberry Ridge tuff, the most widespread (initially covering an area of 15,500 km²) and greatest in volume (1200 km³) of the three tuffs, erupted along circular ring fracture vents extending from Island Park into central Yellowstone National Park. It is a compound cooling unit consisting of three members, each of which is commonly separated from adjacent members by thin ash beds. Dense black vitrophyre occurs at the base of the lower member except at its outer margins, and it may occur at the base of the middle member. The top of the middle member contains a zone of dark and light crystallized pumice fragments and coarse phenocrysts that serves as a useful marker horizon. The upper member, characterized by a pronounced lineation of stretched pumices and aligned phenocrysts, occurs only in the southern part and to the south and southwest of Yellowstone National Park. The upper member locally forms a separate cooling unit, so that

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## TABLE 1. Summary of Microscopic Data From Tuffs of the Yellowstone Group

| Specimen | Locality | R-S | Max | Ilm Lam | Mt Alt | Grdms Red | Grdms Yellow | Red Sil | Fine Opaque | Stability | Remarks |
|---|---|---|---|---|---|---|---|---|---|---|---|
| LAVA CREEK TUFF | | | | | | | | | | | |
| 3YR-152 | Macks Inn | R | 400 | M | MH | MH | 0 | M | H | Stable | Distal end tuff |
| 2YR-93 | Purple Mountains | S | | M | H | | | H | | Stable | Pumice breccia, partially welded |
| 2YR-97-3 | Sheepeater Cliff | S | 300 | | MH | | | | | Stable | Upper vapor phase, partially welded |
| 2YR-90-1 | Lewis Canyon | S | 300 | H | LH | | | | | Stable | Basal ash |
| 0YR-11A | Mt. Everts | S | 220 | 0 | MH | | | MH | | Stable | Basal ash |
| 0YR-12E | Grand Canyon | S | 180 | 0 | L | | | 0 | | Stable | Basal ash |
| 3YR-148 | Tower Falls | R | 300 | OM | L | L | 0 | L | H | Stable | Basal vitrophyre |
| 2YR-94-2 | Sheepeater Cliff | R | 120 | MH | MH | M | L | | H | Stable | Basal vitrophyre |
| 2YR-50-3 | Flagg Ranch | R | 160 | LH | L | L | 0 | L | H | Stable | Basal vitrophyre |
| 2YR-51-3 | Flagg Ranch | S | 220 | MH | LH | M | L | LM | | Stable | Vapor phase, partially welded |
| 2YR-52-10 | Flagg Ranch | R | 220 | LM | 0 | L | 0 | L | H | Stable | Densely welded, devitrified |
| 2YR-65-14 | Grassy Lake | R | 30 | OL | 0 | OL | OL | 0 | H | Stable | Devitrified interior |
| 3YR-144-13 | Virginia Cascades | R | 200 | M | LM | LM | LM | LH | H | Stable | Devitrified interior |
| 2YR-106-2 | Virginia Cascades | R | 140 | LM | LM | L | 0 | 0 | H | Stable | Devitrified |
| MESA FALLS TUFF | | | | | | | | | | | |
| 2YR-108-4 | Island Park Res | R | 400 | L | L | OL | 0 | 0 | H | Stable | Devitrified interior |
| 1YR-32 | Type section | S | 260 | OL | LH | OH | | 0 | | Stable | Pumice in basal ash |
| 1YR-27-11 | Mesa Falls | R | 100 | OL | OL | OL | 0 | 0 | H | Stable | Devitrified interior |
| 2YR-122 | Rte 191 roadcuts | S | 250 | MH | LH | H | MH | H | | Stable | Partially welded, devitrified |
| HUCKLEBERRY RIDGE TUFF | | | | | | | | | | | |
| 2YR-113 | Ririe Dam | R | 140 | MH | M | H | 0 | H | H | Stable | Partially welded, distal end tuff |
| 2YR-111-2 | Ririe Dam | R | 100 | H | MH | H | 0 | H | 0 | Stable | Basalt inclusion |
| 2YR-127 | Casey Lake | R | 120 | MH | MH | H | 0 | H | H | Interm | Partially welded, distal end tuff |
| 2YR-88-4 | Deadman Bar | R | 240 | H | MH | H | L | MH | H | Stable | Distal end tuff |
| 2YR-119-3 | Warm River | S | 210 | MH | OH | | | MH | | Stable | Densely welded, devitrified |
| 2YR-143-1 | Badger Creek | R | 360 | LM | L | LM | L | LM | H | Interm | Devitrified interior |
| 2YR-139-4 | Crown Butte | R | 200 | LH | OH | MH | 0 | MH | H | Stable | Partially welded |
| 2YR-139-6 | Crown Butte | R | 200 | LH | OM | LM | 0 | OM | H | Stable | Densely welded |
| 2YR-123-7 | Lionhead | R | 320 | L | OL | 0 | 0 | 0 | H | Stable | Vitric tuff, middle member |
| 0YR-10G | Mt. Everts | S | 300 | MH | OM | | | | | Interm | Fused basal ash |
| 0YR-10D | Mt. Everts | S | 280 | 0 | OL | | | | | Interm | Basal ash |
| 1YR-16G | Mt. Everts | R | 220 | MH | OL | OL | 0 | OM | L | Stable | Fused basal ash |
| 2YR-140 | Terrace Mountain | R | 200 | LH | OL | L | 0 | LH | H | Stable | Basal vitrophyre |
| 2YR-42-11 | Golden Gate | R | 300 | * | H | H | H | MH | M | Unstable | Devitrified interior |
| 2YR-42-28 | Golden Gate | R | 300 | * | MH | H | M | H | M | Unstable | Devitrified interior |
| 2YR-43-2 | Golden Gate | R | 160 | * | MH | H | M | | H | Interm | Devitrified interior |
| 2YR-44-1 | Golden Gate | R | 120 | LH | LH | MH | MH | M | H | Interm | Vapor phase, lower member |
| 2YR-44-15 | Golden Gate | S | 260 | LM | H | | | H | | | Parting ash between lower-middle members |
| 2YR-45-4 | Golden Gate | R | 240 | 0 | H | H | LM | M | H | Stable | Devitrified, base of middle member |
| 2YR-45-24 | Golden Gate | R | 300 | L | LH | M | L | H | H | Interm | Devitrified interior |
| 1YR-34-5 | Type section | S | 300 | OL | MH | | | | | | Basal ash |
| 2YR-54-2 | Type section | R | 440 | OM | LH | 0 | 0 | 0 | H | Stable | Basal vitrophyre |
| 2YR-55-17 | Type section | R | 110 | OL | M | M | OL | M | H | Interm | Devitrified interior |
| 2YR-56-4 | Type section | R | 220 | OM | M | M | L | M | H | Interm | Devitrified, top of lower member |
| 2YR-57-1 | Type section | R | 120 | 0 | L | M | M | M | H | Interm | Devitrified, base of middle member |
| 2YR-59-15 | Type section | R | 200 | LH | L | L | 0 | | H | Stable | Devitrified, near top of middle member |
| 1YR-37-1 | Type section | S | 200 | MH | MH | | | | | | Parting ash; base of upper member |
| 2YR-60-2 | Type section | R | 140 | LH | LM | MH | L | MH | H | Interm | Devitrified interior |
| 2YR-61-12 | Lizard Creek | R | 260 | LM | LM | LM | L | M | H | Stable | Densely welded |
| 2YR-83-3 | Huckleberry Mtn | R | 200 | M | OL | M | L | M | M | | Basal ash, upper member |
| 2YR-81 | Huckleberry Mtn | S | 240 | OL | OL | | | | | | Basal ash, middle member |
| 2YR-63-4 | Huckleberry Mtn | R | 180 | LM | OL | LM | L | M | H | Stable | Fused basal ash, upper member |
| 2YR-63-15 | Huckleberry Mtn | R | 100 | H | 0 | M | M | M | H | IRM | Devitrified tuff, upper member |
| 2YR-63-3 | Huckleberry Mtn | S | 140 | MH | M | | | | | Stable | Basal ash, upper member |
| 2YR-47-5 | Grayling Creek | S | 300 | OM | L | | | | | Stable | Basal vitrophyre |
| 3YR-145 | Jones Creek | R | 280 | L | OL | L | L | L | H | Stable | Tuff of Notch Mountain Densely welded, devitrified tuff |

R-S: R: polished section of rock
S: polished section of heavy mineral separate

Max: maximum size (in μm) of magnetite phenocrysts

Ilm Lam: presence and amount of ilmenite lamellae in magnetite phenocrysts:
0 not present
L rare
M moderately abundant
H very abundant
* too much low-temperature oxidation to determine presence and degree of high-temperature oxidation

Mt Alt: presence and degree of low-temperature alteration (hematization and maghemitization) of magnetite phenocrysts:
0 very fresh
L fresh to partially altered
M moderately altered
H highly altered

Grdms Red: presence and degree of hematization in ground mass as indicated by red internal reflections (may be deuteric or secondary):
0 no reddening observed
L some reddening of ground mass
M reddening common, especially surrounding altering magnetite and silicate phenocrysts
H extensive reddening

Grdms Yellow: presence and degree of yellow (goethite?) stain in ground mass:
0 - L - M - H same as above, for yellow coloration

Red Sil: presence and degree of alteration (hematization of iron-bearing silicates) 0 - L - M - H same as above for reddening of silicates

Fine Opaque: abundance of tiny opaque particles (≤ 1 μm)
0 not present
L rare
M moderately abundant
H extremely abundant

Stability: stability of specimen or group of specimens against AF demagnetization. (See text) Interm = intermediate stability; IRM: lightning struck.

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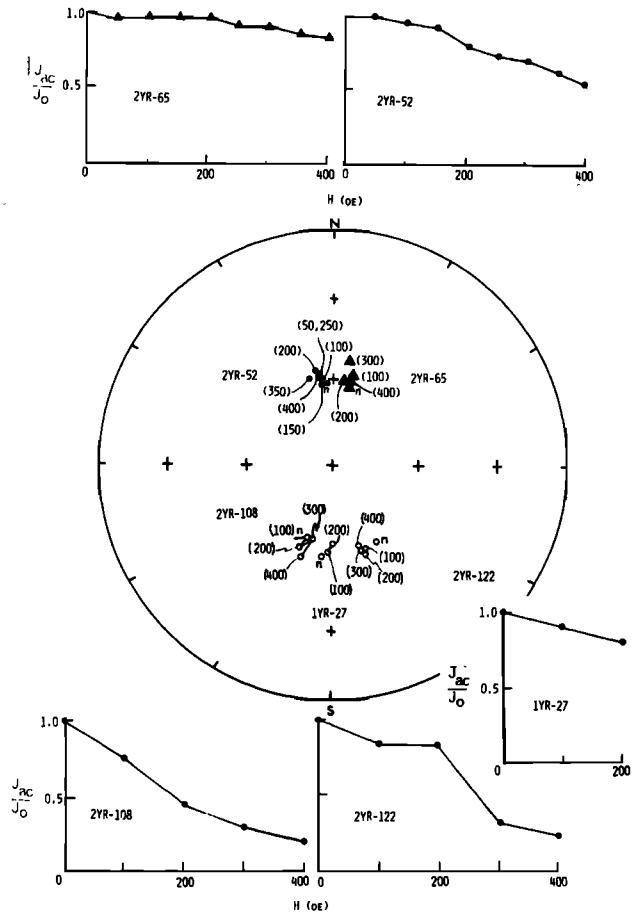



Fig. 2. Alternating field demagnetization of two specimens of Lava Creek tuff and three of Mesa Falls tuff that show little change in direction in response to the demagnetizing field (values in parentheses are in oersteds). Here n refers to NRM. Solid symbols are plotted on lower hemisphere; open symbols are plotted on upper hemisphere of equal-area net. Demagnetization curves plot relative intensity $J_{ac}/J_0$ against demagnetizing field $H$ in oersteds.

the Huckleberry Ridge tuff can be considered a composite sheet.

The Mesa Falls tuff, on the other hand, is a simple cooling unit that originated from the Henrys Fork caldera in the vicinity of Island Park, Idaho. The youngest of the Yellowstone tuffs, the Lava Creek tuff, initially covering more than 7500 km², is a compound cooling unit. It consists of two members that are normally separated by air fall ash, which is indicative of a time interval between eruptive events. Eruption of the Lava Creek tuff created the Yellowstone caldera, which underlies the central portion of the rhyolite plateau.

These distinguishing features, however, are not found in all exposures of the Yellowstone Group. Portable flux-gate magnetometer reconnaissance by Christiansen and Blank indicated that paleomagnetism would be a valuable tool for correlating and distinguishing between the three welded tuffs. They found that the Lava Creek tuff was normally magnetized, whereas the Mesa Falls tuff was reversely magnetized. The Huckleberry Ridge tuff was thought to be magnetized in a horizontal direction in the lower and upper members but normally magnetized in the middle member. This study is in part an extension of their efforts as an aid to the geological mapping of the Yellowstone rhyolite plateau.

## PALEOMAGNETIC PROCEDURES

Samples for paleomagnetic analysis were either taken as large hand samples or drilled from the outcrop (as 2.54-cm cylinders) by means of a portable gasoline rock drill and oriented by means of a Brunton compass. In the laboratory, 2.54-cm-diameter specimens were drilled from the hand samples. The natural remanent magnetization (NRM) of all specimens (approximately 1000) was first measured on a 10-Hz Marathon spinner magnetometer with a sensitivity of about 5 $\times 10^{-5}$ emu/cm³. Following that, about 800 specimens were demagnetized by alternating field (af) methods in a three-axis tumbler in peak fields up to 400 Oe. Partial thermal demagnetization was performed on 300 specimens, two thirds of which had not been previously demagnetized by af methods. Most specimens were heated in air in a noninductively wound furnace enclosed in a $\mu$-metal shield in which the residual magnetic field was $\pm 4$ $\gamma$. Thermal demagnetization of specimens used in making determinations of the intensity of the ancient magnetic field was done in a furnace in which the atmosphere could be controlled by mixing $H_2$ and $CO_2$ [*Larson et al.*, 1975]. After thermal demagnetization the remanence of all specimens too weak to be measured on the slow speed magnetometer was measured on an air turbine (155 Hz) Marathon spinner magnetometer with a sensitivity of about $2 \times 10^{-7}$ emu/cm³.

Reflection microscopy of polished sections at magnifications up to $\times 1200$ was used to help identify opaque oxides. In addition, it was used to assess grain sizes and high- and low-temperature oxidation states of magnetic grains, information

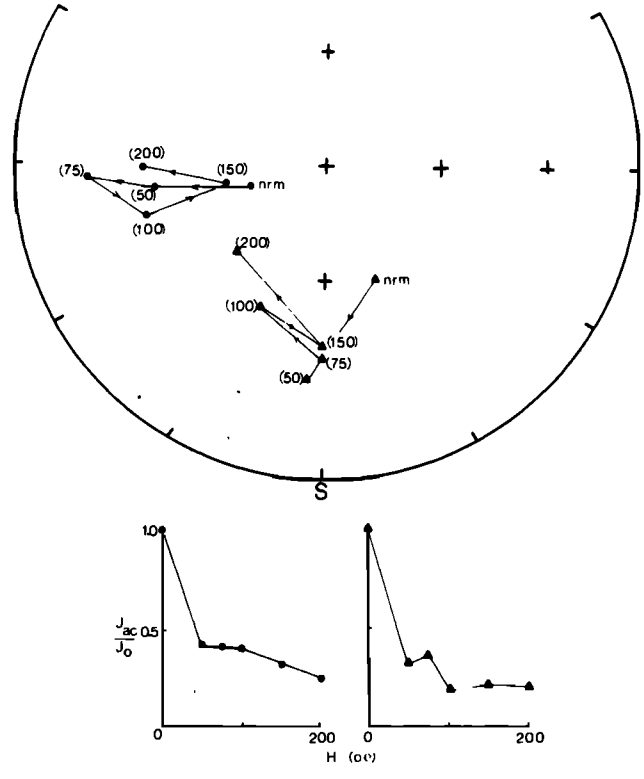



Fig. 3. Alternating field demagnetization of two specimens (from the same hand sample) of devitrified tuff from the interior of the Huckleberry Ridge tuff (lower member) at Golden Gate (site 2YR-42) showing large changes in remanence directions and failure of directions to converge at any cleaning level. The notation is the same as that used in Figure 2.

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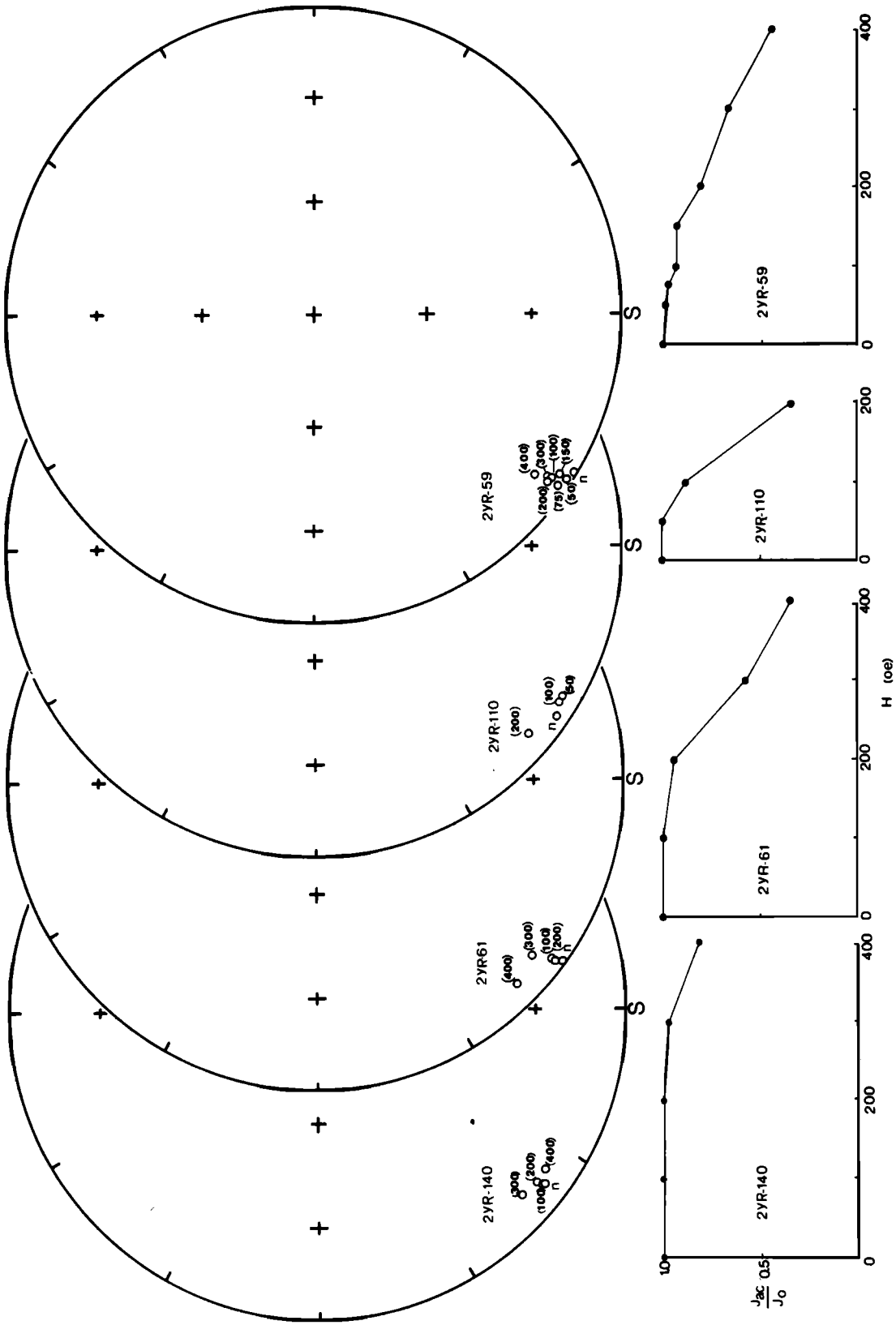



Fig. 4. Alternating field demagnetization of four specimens of Huckleberry Ridge tuff that show little directional change in response to demagnetization. The notation is the same as that used in Figure 2.

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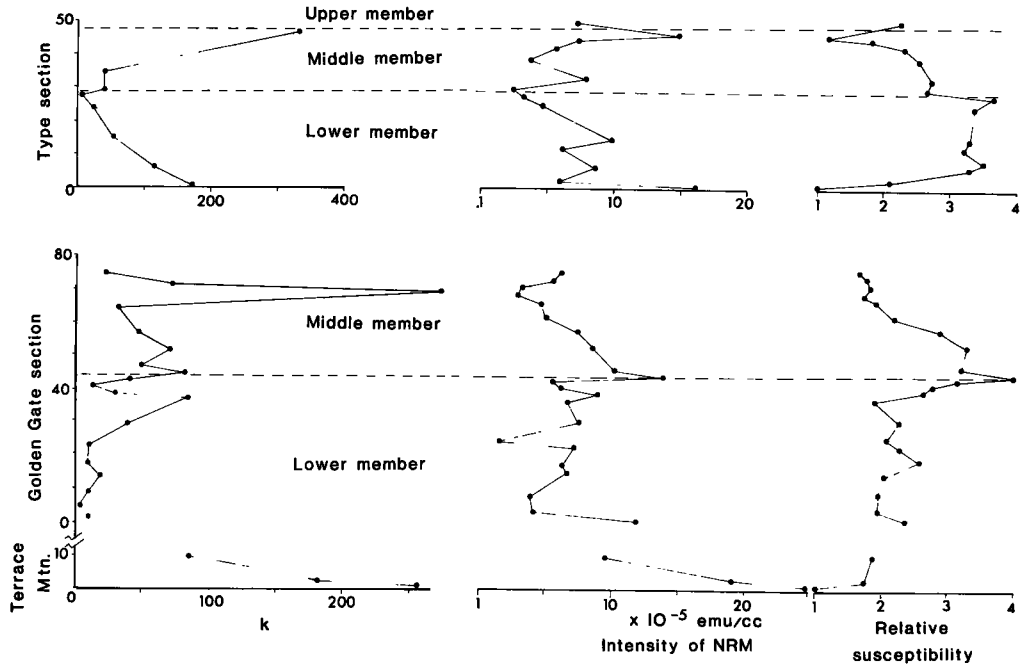



Fig. 5. Variation of $k$, intensity of NRM, and relative susceptibility with stratigraphic position for the Golden Gate section and type section of the Huckleberry Ridge tuff. Values of $k$ are for groups ($N \geq 4$) of specimens giving least dispersion following af demagnetization. Intensities of NRM and relative susceptibilities are for selected specimens. Relative susceptibility values were normalized to values obtained from the basal vitrophyre. The stratigraphic distance is in meters above the base of the section.

that bears directly on conclusions concerning the stability and reliability of remanence directions. Compositions of selected magnetite grains were determined by electron microprobe.

Thermomagnetic analysis, saturation moment $J_s$ versus temperature $T$, made in a field of 2000 Oe on a Cahn electrobalance was also used in the identification of the magnetic iron–titanium oxide minerals. The Curie balance used in the $J_s$-$T$ analysis also has gas ($H_2$ and $CO_2$) mixing capabilities and a means of continuously monitoring the oxygen fugacity [*Larson et al.*, 1975]. The maintenance of an appropriate oxygen fugacity during heating greatly retards oxidation of titanomagnetite (to magnetite and ilmenite) and magnetite (to hematite). It is also possible by gas-controlled $J_s$-$T$ experiments to identify secondary magnetic components such as maghemite and titanomaghemite, which upon heating will break down to hematite and magnetite, respectively, regardless of atmosphere.

## Stability of Remanent Magnetization and Rock Magnetic Properties

### *Stability of Remanence*

Stability of remanence to af demagnetization for specimens or groups of specimens for which polished sections were examined is given in Table 1. 'Stable' indicates that directions following demagnetization at all cleaning levels were closely similar ($\leq 10°$) to NRM directions. 'Intermediate' stability describes the case in which directions of NRM and directions after demagnetization in low fields (at 20 or 50 Oe) differed substantially ($>10°$) from directions attained after application of higher demagnetizing fields and the directions after cleaning to progressively higher fields reached a stable position, defined as lying within a cone of 5° radius. 'Unstable' indicates that individual specimens greatly changed in direction from NRM through all stages of demagnetization, never reaching a stationary direction, and that adjacent specimens and/or specimens from the same core possessed divergent directions of NRM which failed to group closely after cleaning, regardless of the level of the demagnetizing field.

All samples of the Lava Creek and Mesa Falls tuffs are stably magnetized, as is indicated by both thorough af and thermal demagnetization tests (Table 1 and Figure 2). In contrast, most of the Huckleberry Ridge tuff is magnetically unstable or of intermediate stability (Table 1) in response to af demagnetization (Figure 3). NRM directions for many of these specimens appear to have been affected by remagnetization in a field of normal polarity (probably within the last 700,000 yr), but much of the normal component can be removed by af cleaning in relatively low fields. Consequently, af demagnetization induces large changes in direction as the re-

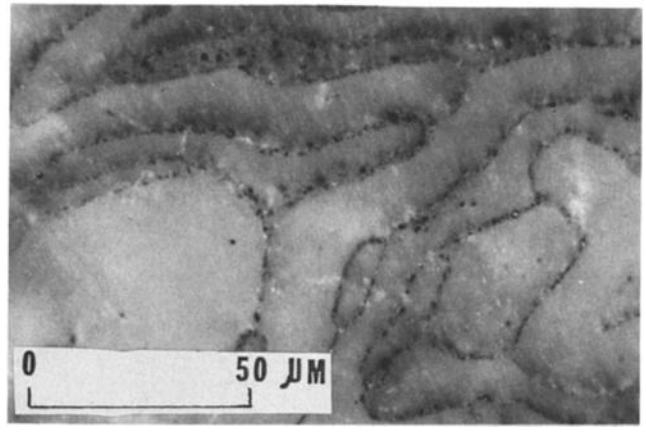



Fig. 6. Photomicrograph of densely welded Huckleberry Ridge tuff showing concentrations of opaques (1- to 2-$\mu$m diameters) at the margins of deformed glass shards.

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TABLE 2. Electron Microprobe Data of Magnetite Phenocrysts for Yellowstone Group

| Unit & Sample | Magnetite* Fe | Ti | Mg | N | Ulvospinel Mol % | Remarks |
|---|---|---|---|---|---|---|
| Lava Creek Tuff | | | | | | |
| 2YR-51-3 | 57.0 | 11.3 | .11 | 13 | 56 | Upper vapor phase; lower member |
| 3YR-148 | 58.5 | 10.5 | 1 | 4 | 52 | Basal vitrophyre; upper member |
| 2YR-106-2 | 57.2 | 11.3 | .08 | 1 | 56 | Devitrified base; upper member |
| 2YR-90-1 | 56.5 | 11.0 | .28 | 15 | 54 | Non-welded basal ash; upper member |
| Mesa Falls Tuff | | | | | | |
| 1YR-32 | 57.7 | 10.8 | .11 | 15 | 53 | Pumice; basal air fall |
| Huckleberry Ridge Tuff | | | | | | |
| 2YR-34-5 | 58.5 | 11.9 | .09 | 15 | 57 | Non-welded ash flow below basal vitrophyre; type section |
| 2YR-54 | 57.5 | 11.4 | .1 | 12 | 56 | Basal vitrophyre; lower member type section |
| 2YR-47-5 | 56.3 | 12.3 | .1 | 14 | 60 | Basal vitrophyre; lower(?) member |
| 0YR-10D | 59.2 | 10.0 | .02 | 15 | 49 | Basal air fall ash; lower member |
| 0YR-10G** | 54.5 | 11.6 | .03 | 15 | 59 | Fused air fall ash; 2 m above 0YR-10D |
| 2YR-81 | 59.5 | 11.0 | .08 | 15 | 53 | Basal air fall ash; middle member |

*Values in weight percent.
**Abundant ilmenite lamellae.
N - Number of grains analyzed.

sulting directions become shallower in inclination and move toward the stable remanence directions as exhibited in other portions of the tuff. Alternating field cleaning of specimens most affected by remagnetization, however, fails to achieve close grouping in the subhorizontal southwest direction. Stably magnetized rocks of the Huckleberry Ridge tuff are primarily limited to the base and densely welded margins of partial cooling units and the partially welded tuff of the distal margins (Table 1 and Figure 4).

The precision parameter $k$ for groups of specimens, intensities of NRM, and relative susceptibilities, as they vary with vertical distribution in thick sections, reflect the variations observed in the stability of the Huckleberry Ridge tuff (Figure 5). Values of $k$ and intensities of NRM are both greater for stably magnetized rocks of the basal vitrophyre and overlying spherulitic tuff than for the predominately unstable rocks of the flow interior. Susceptibility is relatively higher in the interiors of the sheets than at the margins. High $k$ values and NRM intensities are also seen for the middle member in the lower margin at Golden Gate and in the upper margin at the type section, but this correspondence is not consistent throughout the interiors of the sheets.

The grain sizes and compositions of the primary magnetic phases, their thermomagnetic characteristics as reflecting high-temperature oxidation, and properties of secondary magnetic phases were studied in an attempt to identify the sources of stable and unstable magnetism in the Yellowstone tuffs. Pertinent observations are discussed in the following sections.

### *Grain Size of Magnetic Minerals*

Magnetite and titanomagnetite grains range in size from about 400 to less than 1 $\mu$m. Grains of very fine particle size (less than 2 $\mu$m) are ubiquitous (Figure 6), and they probably carry most of the stable remanence observed in the three tuffs [see *Ozima and Ozima,* 1965; *Verhoogen,* 1959; *Larson et al.,* 1969]. All the observable tiny opaques are probably of pseudo-single-domain (PSD) size [see *Dunlop,* 1973], but although single-domain (SD) grains are not distinguishable with a reflecting microscope, they are likely to occur in these tuffs. Also present but rare are magnetite and titanomagnetite phenocrysts with a bimodal size distribution (averaging about 200 and 15 $\mu$m). Many of these phenocrysts have undergone high-temperature oxidation to produce ilmenite 'exsolution'

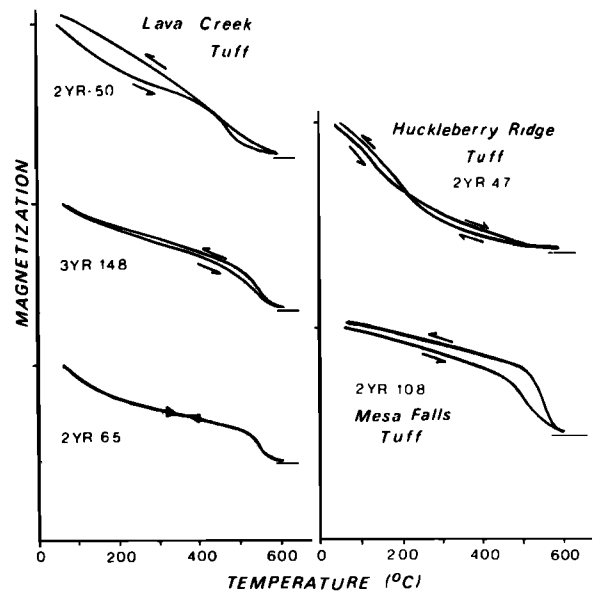



Fig. 7. Thermomagnetic curves of basal vitrophyre (samples 2YR-65, 2YR-50, 2YR-47, and 3YR-148) and devitrified tuff (sample 2YR-108) indicating the presence of titanomagnetite. Samples 2YR-65, 3YR-148, 2YR-47, and 2YR-108 were heated in $H_2 + CO_2$; sample 2YR-50 was heated in air.

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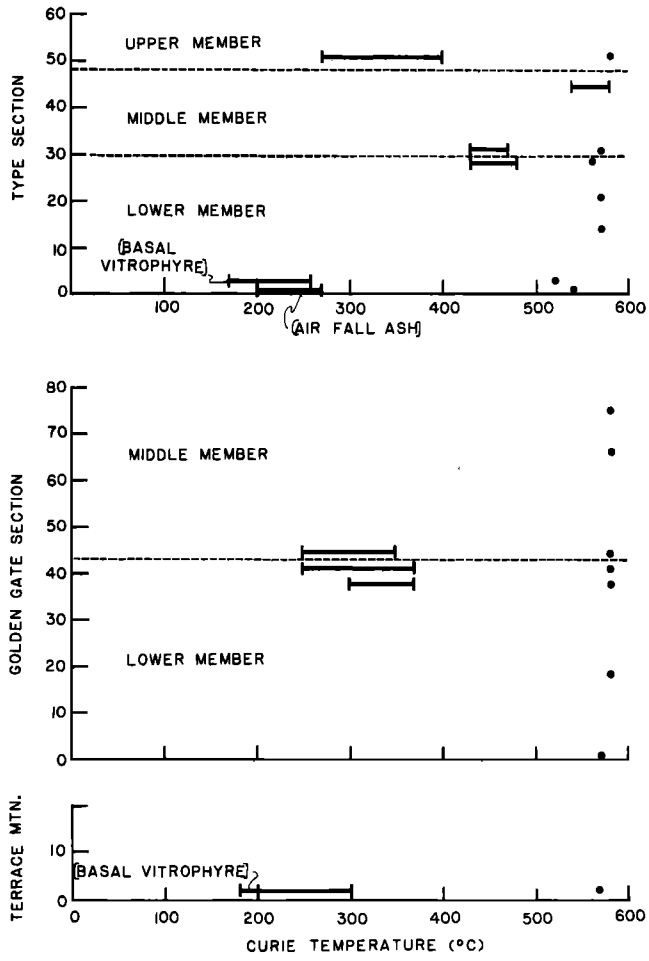



Fig. 8. Variation of Curie temperatures with vertical position in the Huckleberry Ridge tuff. Solid circles indicate discrete Curie temperatures; bars represent a range of Curie temperatures. All samples having titanomagnetite with a range of low Curie temperatures also contained magnetite with Curie points near 580°C. The stratigraphic distance is in meters above the base of the section.

lamellae along (111) planes. This phenomenon can result in the reduction of the overall grain size into PSD and SD size segments, thereby greatly increasing the magnetic stability of the rocks, as has been documented for basalts [*Larson et al.*, 1969]. Titanomagnetites subdivided by ilmenite lamellae may partly contribute to magnetic stability in the Yellowstone tuffs, but there is no firm correlation between high-temperature oxidation state (development of ilmenite lamellae) and magnetic stability (Table 1). Although some specimens in which magnetite is heavily laced with ilmenite lamellae are stably magnetized, others are not. Stable magnetism also resides in samples not containing exsolved ilmenite in the large titanomagnetite grains.

### *Composition of Magnetic Minerals and High-Temperature Oxidation*

Most $J_s$-$T$ curves for the Yellowstone tuffs show Curie points of 560°–580°C, indicating that magnetite with little or no Ti content is the dominant phase in the samples. The original composition of magnetic minerals, however, was enriched in ulvospinel. Titanomagnetites (134 grains from 11 samples of pumice, basal ash, vitrophyre, or otherwise rapidly cooled tuff) were analyzed by microprobe and found to range from 49 to 60 mol % and average 55 mol % ulvospinel (Table 2). Titaniferous magnetites of about 55 mol % ulvospinel can be expected to have a Curie temperature near 250°C [*Readman and O'Reilly*, 1972], but few samples of tuff from the Yellowstone Group demonstrated this single discrete Curie temperature. Nonetheless, many $J_s$-$T$ curves of samples, including those used for microprobe study, show inflection between about 200° and 450°C (Figure 7), which may be indicative of a range of titanomagnetite compositions resulting from varying degrees of high-temperature oxidation. As high-temperature oxidation proceeds, titanomagnetite is oxidized to ilmenite, which diffuses out of the spinel structure, and to magnetite which results in an increase in the Curie temperature.

The presence of magnetite and titanomagnetite reflecting different degrees of high-temperature oxidation varies systematically with position in the flows but without any meaningful correspondence to variations in magnetic stability. High-temperature oxidation (most likely a deuteric alteration) has been greatest toward the center of the ash flow sheets, as is indicated by the predominance of pure magnetite in the interior of the partial cooling units and of titanomagnetite at the more rapidly cooled base and margins of partial cooling boundaries (Figure 8).

Pure magnetite is also characteristic of tuff of the distal edges (Figure 9). Apparently, oxidizing conditions were more favorable in these tuffs than in the rapidly chilled portions of tuff nearer to their source. Perhaps the longer duration of flow for distal end tuff than for near-source tuff and the greater incorporation of air during flow could have promoted oxidizing conditions in the outer margins of the ash flow sheet.

### *Low-Temperature Oxidation*

Hematite, goethite, other ferric oxide phases, and maghemite occur in the Yellowstone tuffs as secondary magnetic phases formed by low-temperature oxidation of preexisting iron–titanium oxide minerals and other iron-bearing grains. Hematite (with a Néel temperature of about 670°–680°C) has been identified in many samples by thermomagnetic analysis. In addition, magnetic phases with Curie or Néel points in the range between 580° and about 650°C are commonly present and represent either titanohematite of very high $Fe_2O_3$ mole percent or poorly crystallized hematite with a range of Néel temperatures up to about 680°C. It is impossible, of course, to determine solely by thermomagnetic analysis whether these hematitic phases are deuteric (from the high-temperature oxidation of magnetite) or low-temperature alteration products.

Polished section study is capable of giving some indication of the origin of the hematite. Hematite seen as internal reflections is common in many samples as pigment within the groundmass, especially adjacent to rock fractures, and surrounding altered magnetites and iron-bearing silicates (Table 1). In many instances these silicates are extremely altered and reddened, a feature reminiscent of the red clastic deposits described by *Walker* [1967] and *Walker et al.* [1967]. Undoubtedly, these diagenetic alterations, the liberation and reprecipitation of iron (as $\alpha$-$Fe_2O_3$) by intrastratal solutions, are potentially important contributors to secondary remanence in these pyroclastic deposits.

Thermal demagnetization is also useful for distinguishing between primary and secondary hematite in the Mesa Falls and Huckleberry Ridge tuffs. Heating to 600°C and above (that is, above the Curie temperature of $Fe_3O_4$) and cooling in field free space isolate a reversed direction identical to directions obtained after af demagnetization for the three localities

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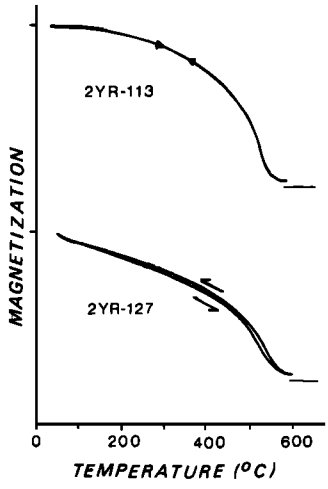



Fig. 9. Thermomagnetic curves of distal end tuff of the Huckleberry Ridge tuff. Sample 2YR-113 had the same $J_s$-$T$ curve regardless of the heating environment ($H_2$ + $CO_2$ or air); sample 2YR-127 was heated in $H_2$ + $CO_2$.

of the Mesa Falls tuff. Remanence directions after similar heatings of samples of the Huckleberry Ridge tuff from several localities are essentially unchanged from directions of stable NRM or directions after af cleaning, indicating that either titanohematite or hematite carries some of the stable remanence parallel to the original direction in these samples. Although thermal demagnetization at temperatures greater than 600°C of certain samples of Lava Creek tuff leads to the isolation of a normal direction of magnetization identical to the NRM, it is impossible to know if this direction represents a partial thermochemical remanent magnetization (PTCRM) acquired at temperatures greater than 600°C or is a low-temperature chemical remanent magnetization (CRM).

In contrast, secondary hematite is present in samples of the Huckleberry Ridge tuff near Casey Lake, Montana (site 2YR-127), which exhibit reddening of silicates and groundmass in polished section. Thermal demagnetization to 500°C causes the specimens to shallow in inclination toward the original direction of thermal remanent magnetization (TRM) for the unit, but further heating to 565°C causes steepening of the inclinations, and the directions migrate to the north (Figure 10). Further demagnetization at 600° and 630°C isolates a normal direction of magnetization presumably acquired during the last 700,000 yr.

Goethite, whose presence is suggested by yellow staining of rocks in outcrop, also appears to contribute to the chemical overprinting of the original magnetization of the Huckleberry Ridge tuff. Yellow discoloration is especially marked in the devitrified tuff of the Golden Gate section, for which overprinting in a normal direction is most clearly seen. Thermal cleaning of these samples at 90°C, which is below the 120°C Néel point of goethite given by *Strangway et al.* [1968], gives a mean direction ($D$ = 239°, $I$ = 46°, and $\alpha_{95}$ = 15°) not greatly different from the direction of NRM ($D$ = 223°, $I$ = 43°, and $\alpha_{95}$ = 17°). Demagnetization at 180°C, however, greatly reduces dispersion and isolates a mean ($D$ = 222°, $I$ = 7°, and $\alpha_{95}$ = 8°) near the established mean direction of this unit (see the section on paleomagnetism of the Huckleberry Ridge tuff). The goethite may not be entirely related to surficial weathering processes; it may also be associated with alteration by hydrothermal fluids because goethite occurrences are located near areas of hot springs activity. The remanence in the goethite may then be a postdepositional TRM (or PTCRM) if introducing waters are near 120°C.

In contrast to the Huckleberry Ridge tuff at Golden Gate, yellow pigmentation is not observed in outcrop and is not greatly evident in polished sections of samples from the type section (Table 1). Thermal cleaning at 100° and 200°C of specimens with initially steep inclinations improves the NRM by beginning to isolate the original TRM but does not remove the secondary overprints as thoroughly as it does for the Golden Gate samples. This is seen in many samples from other sites as well. The hematite with high blocking temperatures and high coercivities and the goethite, then, are not the only phases contributing to the remagnetization of the Huckleberry Ridge tuff in a normal direction. Secondary magnetic phases, perhaps ferric hydroxide phases other than goethite, with a range of low coercivitives and low blocking temperatures could also be important in the development of chemical remagnetization.

Maghemite and titanomaghemite (low-temperature oxidation products of magnetite and titanomagnetite, respectively) have also been identified in thermomagnetic curves of many samples and observed petrographically as replacements of host phenocrysts along the grain margins and adjacent to fractures within the grains. It could not be determined petrographically if the very small (<2 μm) magnetite particles had been partly or completely affected by maghemitization. Because the size and shape characteristics of maghemite in the Yellowstone tuffs are incompletely understood, it is difficult to assess directly the role of maghemite as a possible source of magnetic instability. Nevertheless, there is no apparent correlation between the presence of maghemite and unstable remanence. For example, wherever the Lava Creek tuff is sampled, it is magnet-

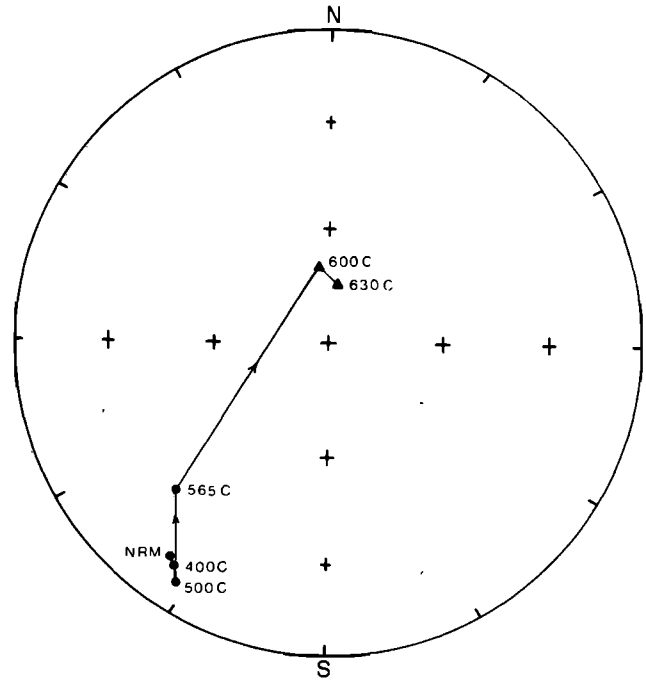



Fig. 10. Thermal demagnetization of specimens of the Huckleberry Ridge tuff at Casey Lake, Montana (sample 2YR-127). Circles indicate the mean of two specimens; triangles show the direction of a single specimen demagnetized at 600° and 630°C.

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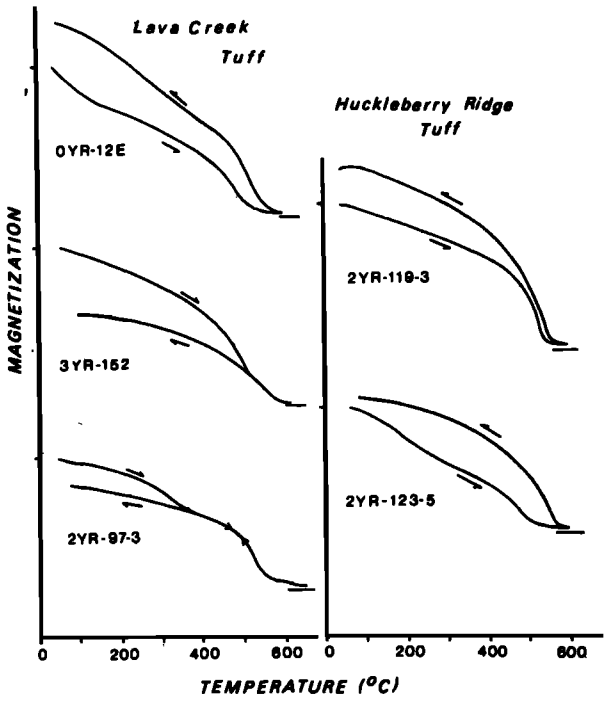



Fig. 11. Thermomagnetic curves of specimens which are magnetically stable but which have undergone maghemitization (either maghemite or titanomaghemite). All samples were run in $H_2 + CO_2$ and separately in air.

ically stable, and maghemite has been identified by means of $J_s$-$T$ curves (Figure 11) and polished sections in rocks from at least three collecting sites in that unit. Although portions of the Huckleberry Ridge tuff that are of intermediate magnetic stability contain maghemite (for example, within the Golden Gate and type sections), other parts that are stable also contain titanomaghemite or maghemite (for example, sites 2YR-54, 2YR-123, and 2YR-119 and the lowermost tuff of the middle member at Golden Gate (Table 1 and Figure 11)). Furthermore, rocks of intermediate stability at sites 2YR-60 and 2YR-127 contain no maghemite.

## Origin of Magnetic Instability in the Huckleberry Ridge Tuff

As is indicated above, large homogeneous grains and maghemite are not solely responsible for the apparent magnetic instability of samples of the Huckleberry Ridge tuff. Rather, magnetic instability appears primarily to be related to a combination of different magnetic phases carrying different directions of magnetization and possessing highly variable ranges of blocking temperatures and coercivities. Removal by demagnetization of a secondary overprint in a normal direction that is carried by magnetic phases with low coercivities and low blocking temperatures causes large changes in remanence directions as the resulting directions move toward the direction of the TRM. If no other secondary magnetic minerals are present, the cleaned directions remain essentially stationary about the mean direction of TRM. Yet when two or more secondary phases with a broad range of coercivities are present in a sample, further changes in direction occur with the application of higher alternating fields. This is illustrated in Figure 12, where cleaning in fields less than 200 Oe removes a secondary overprint and begins to uncover the original TRM; however, demagnetization at 200 Oe causes the direction of the sample to migrate to the north beyond the direction of NRM. With further cleaning up to 400 Oe the directions continue to steepen and move northward toward a normal direction. In samples such as this, alternating fields remove magnetization carried in grains of relatively low coercivities and thereby isolate a secondary normal component of remanence carried in hematite with higher coercivities.

Partial remagnetization, then, is largely responsible for significant changes in remanence directions in response to demagnetization and is most clearly seen in the interior devitrified portions of the flow, where groundwater or hydrothermal solutions have led to the growth of secondary magnetic phases. In contrast, densely welded tuffs (presumably zones of low permeability), especially at the base and margins of partial cooling units, are magnetically stable. Consistent with these observations are preliminary determinations of porosity that suggest correlation of low porosity with stably magnetized rocks and high $k$ values and correlation of high porosity with unstable magnetization and low $k$ values. Two samples of densely welded basal vitrophyre have porosities of 4.1 and 5.0%, which differ greatly from an average porosity of 11.9% (variance of 1.4) for six samples of devitrified tuff that are of intermediate stability. It seems likely that oxidizing fluids promoted remagnetization in rocks of relatively high porosity long after emplacement of the flow but were not as easily introduced into more densely welded tuff of lower porosity. Stable magnetism also resides in tuff at the outer margins of the flow, where groundwater and hydrothermal influences have been minimal.

Although low-temperature oxidation products are present in the Lava Creek and Mesa Falls tuffs, they have not greatly altered the remanence directions. Small amounts of chemical overprinting in these tuffs will not be readily apparent inasmuch as their initial remanence is relatively so strong. The intensity of NRM for the Lava Creek tuff averages $3.1 \times 10^{-3}$

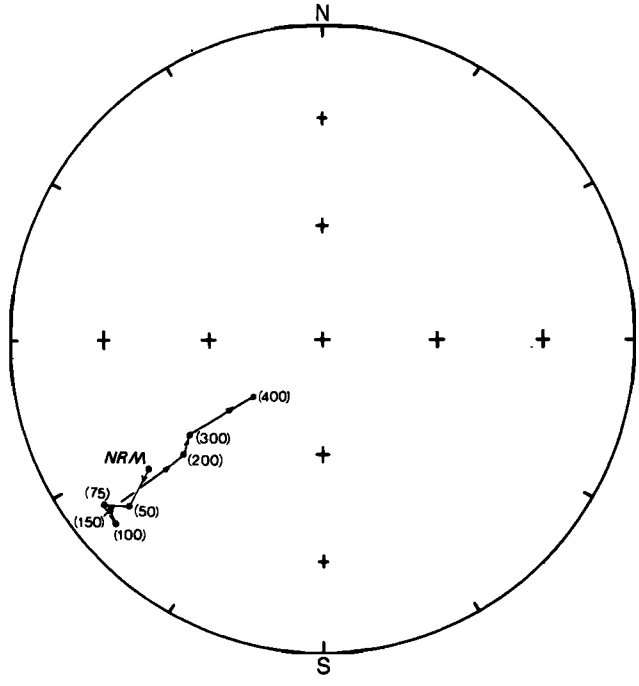



Fig. 12. Alternating field demagnetization of specimen 2R-292 (from the upper member at the type section of Huckleberry Ridge tuff) showing isolation of normal direction with increasing af peak values.

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TABLE 3. Summary of Paleomagnetic Data for Lava Creek Tuff

| Site | Locality | MMB | Remarks | N | $\bar{H}$(oe) | $\bar{I}$ | $\bar{D}$ | k | $\alpha_{95}$ |
|---|---|---|---|---|---|---|---|---|---|
| 2YR-50 | Flagg Ranch | a | VIT | 8 | 100 | 60.6 | 5.6 | 281.2 | 3.3 |
| 2YR-51 | | a | UVP | 3 | 200 | 67.0 | 12.3 | 648.7 | 4.9 |
| 2YR-52 | | b | DVIT | 8 | 150 | 62.8 | 5.9 | 302.9 | 3.2 |
| 3YR-144 | Virginia Cascades | a | DVIT | 8 | 100 | 67.8 | 357.0 | 614.2 | 2.2 |
| 2YR-106 | | b | DVIT | 8 | 150 | 62.0 | 354.2 | 254.3 | 3.5 |
| 2YR-65 | Grassy Lake Quarry | b? | DVIT+SB | 8 | 100 | 61.5 | 2.2 | 64.0 | 6.9 |
| | | b? | VIT | 5 | 200 | 54.2 | 6.0 | 417.3 | 3.8 |
| 3YR-152 | Macks Inn roadcuts | ? | DVIT | 7 | 200 | 68.9 | 1.2 | 552.7 | 2.6 |
| 2YR-93 | Purple Mountain | a* | PB | 6 | 50 | 65.1 | 9.3 | 74.9 | 7.8 |
| | | a | I | 3 | 50 | 65.7 | 5.7 | 117.3 | 11.4 |
| 2YR-94,95 | Sheepeater Cliff | b* | VIT+SB | 6 | 200 | 71.1 | 5.8 | 392.3 | 3.4 |
| 2YR-96,97 | | b | DVIT | 7 | 200 | 69.5 | 7.4 | 46.5 | 8.9 |
| 2YR-90 | Lewis Canyon | b? | FA+SB | 7 | 200 | 60.3 | 359.7 | 36.1 | 10.2 |
| 3YR-148 | Tower Falls roadcut | b | VIT | 4 | 200 | 60.3 | 352.2 | 195.7 | 6.6 |
| | | b | DVIT | 7 | 100 | 62.8 | 14.2 | 217.7 | 4.1 |
| 0YR-13 | Grand Canyon | b | DVIT | 5 | 100 | 67.9 | 7.8 | 434.5 | 3.7 |
| 0YR-12 | Grand Canyon | b | AA | 14 | 50 | 63.8 | 355.5 | 50.1 | 5.7 |
| | | b | FA+VIT | 10 | 100 | 66.8 | 3.4 | 175.2 | 3.7 |
| 2YR-114 | Warm R. source | ? | DVIT | 3 | 100 | 76.5 | 15.9 | 1066.0 | 3.8 |
| 2YR-124 | Grayling Creek | ? | DVIT | 5 | 100 | 66.6 | 354.5 | 63.8 | 9.7 |
| 3YR-154 | Warm River overlook | ? | DVIT | 3 | 100 | 67.0 | 3.1 | 32.6 | 21.9 |
| 2YR-67,68 | Reclamation Road | b? | DVIT | 5 | 100 | 69.4 | 347.4 | 46.1 | 11.4 |
| 2YR-132 | Cache Creek | b? | DVIT | 6 | 200 | 63.7 | 344.2 | 148.7 | 5.5 |
| 2YR-135 | Cave Falls | b? | VIT+DVIT | 4 | 200 | 62.8 | 7.7 | 82.3 | 10.2 |

MMB indicates member of Lava Creek Tuff, a = lower; b = upper; b? where correlation uncertain Query where unknown. a* reference section for lower member; b* reference section for upper member

VIT = basal vitrophyre

UVP = upper vapor phase zone

DVIT = devitrified tuff

SB = densely welded; blue, spherulitic tuff directly above vitrophyre

PB = pumice breccia

I = inclusions

FA = fused ash

AA = air fall ash

S indicates drilled samples (D) or hand samples (H)

N number of specimens

$\bar{H}$ alternating current demagnetizing field in oersteds yielding least dispersion (highest k value)

$\bar{I}$ mean inclination, degrees, positive downward

$\bar{D}$ mean declination, degrees from true north

k precision parameter (Fisher, 1953)

$\alpha_{95}$ semi-angle of the cone of 95 percent level confidence

(2YR-51 and 2YR-93(I) combined with 2YR-50 and 2YR-93(PB), respectively in calculating unit mean.

emu/cm$^3$ for vitrophyre and $1.4 \times 10^{-3}$ emu/cm$^3$ for devitrified tuff, and for the Mesa Falls tuff (all devitrified) it averages $1.0 \times 10^{-3}$ emu/cm$^3$. In contrast, the Huckleberry Ridge tuff, which apparently was erupted and cooled during a time when the earth's field was only about one tenth as strong as it was at 1.2 and 0.6 m.y. B.P. (see the section on paleomagnetism of the Huckleberry Ridge tuff), averages $4.5 \times 10^{-4}$ and $6 \times 10^{-5}$ emu/cm$^3$ for vitrophyre and devitrified tuff, respectively. In the Huckleberry Ridge tuff, then, a relatively weak initial intensity and later oxidation and acquisition of CRM in large fields have caused relatively large changes in remanence direction and intensity.

## PALEOMAGNETISM OF THE LAVA CREEK TUFF

Paleomagnetic determinations of the Lava Creek tuff are based on 157 oriented specimens from 22 collecting sites at 15 separate localities. The mean directions giving least dispersion for each site are listed in Table 3. The unweighted mean direction of the 22-site means is $D = 1.8°$, $I = 65.9°$ ($\alpha_{95} = 2.1°$, $k = 217.4$), yielding a virtual geomagnetic pole (VGP) at 86.1°N, 93.1°W ($\partial p = 2.8°$, $\partial m = 3.4°$).

Comparisons of paleomagnetic directions within sites and between sites of the same member were made to determine directional variation within this young but widespread ash flow sheet. In general, agreement within members is good, but discrepancies do exist. Mean remanence directions of friable air fall ash, fused air fall ash, and basal vitrophyre at locality 0YR-12 are statistically indistinguishable (Table 3). The origin of magnetization of the air fall ash is not known, but it is perhaps a TRM or a PTCRM, the ash having been deposited as hot shards and/or heated above the blocking temperatures of some of the included ferromagnetic grains during sub-

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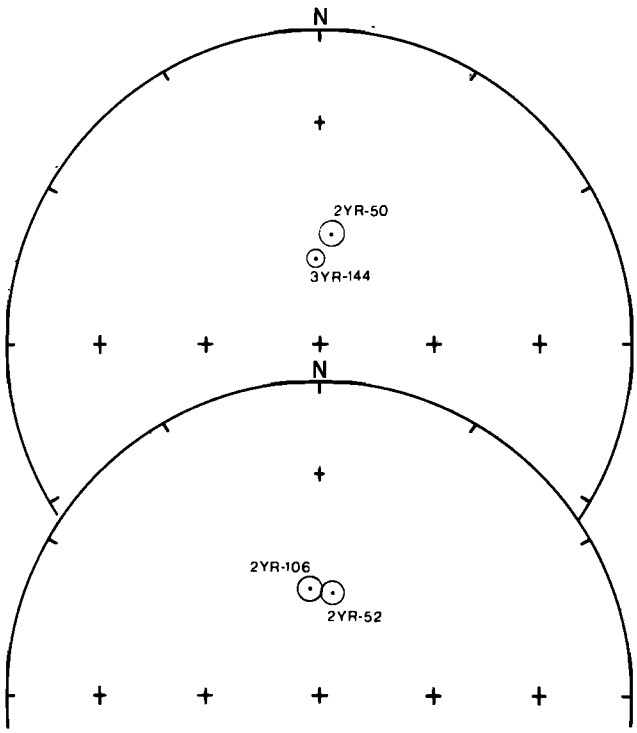



Fig. 13. (Top) Mean directions of remanence of the lower member of the Lava Creek tuff at Virginia Cascades (3YR-144) and at Flagg Ranch (2YR-50). (Bottom) Mean directions of remanence of the upper member of the Lava Creek tuff at Virginia Cascades (2YR-106) and at Flagg Ranch (2YR-52).

sequent emplacement of the overlying ash flow sheet. Additionally, no consistent differences in remanence directions between vitrophyre and overlying devitrified tuff were observed that might suggest a different process or source of magnetization of the two types of tuff.

Among the discrepancies observed are those within the upper and lower members at the Flagg Ranch and Virginia Cascades localities. The cone of 95% confidence level ($\alpha_{95}$) of the lower member at Flagg Ranch (site 2YR-50) does not overlap that of the same member at Virginia Cascades (site 3YR-144, Figure 13), the angular difference between the mean directions being 8°. Furthermore, although the $\alpha_{95}$ of the upper members at these localities (sites 2YR-52 and 2YR-106) intersect, their means are 6° apart and cannot be regarded as being identical according to *Watson*'s [1956] $F$ ratio tests.

Tectonic tilting, differences in cooling rates, flowage-related magnetic anisotropy, and local magnetic anomalies are considered to be potential sources of dispersion within the Lava Creek tuff. Influence from a single source of dispersion, however, appears insufficient to account for all the observed discrepancies within the unit; it is more likely that contributions of varying degrees from each source of error have resulted in the dispersion of remanence directions.

The study area is one of Quaternary tectonic activity, and faulting continues to the present time (for example, the West Yellowstone–Hebgen Lake activity, *Smith and Sbar* [1974]; *Love* [1961]). At localities where planar surfaces of the tuff define a measurable dip, it is difficult to assess whether this dip is caused by tectonic tilting or is a primary compaction dip reflecting uneven preeruption topography. No dips greater than 6° were measured at sites of the Lava Creek tuff, so that differences in mean directions between sites due to tectonic factors should not exceed about 10° at most, which is in excess of most discrepancies measured. Structural corrections were performed for the Cache Creek locality (5°) and the Virginia Cascades sites (4°), where tilting appeared to be continuous throughout the thick section of tuff.

It is also possible that substantial differences in cooling rates may cause the 'freezing in' of different magnetic directions in different parts of tuff sheets as the earth's magnetic field continued to undergo secular variation. However, cooling of the Lava Creek tuff, determined for sampled localities following *Jaeger*'s [1968] models for cooling extrusive rocks, was sufficiently rapid in relation to present rates of secular variation that significant differences ($\geq 10°$) in remanence directions are not solely attributable to effects of different cooling rates.

Preferential alignment of elongate grains causing a particle-induced magnetic anisotropy [*Uyeda et al.*, 1963] might be considered as another source of dispersion. Inasmuch as eruption occurred radially outward from circular vents, the nuée ardente must have flowed in different directions, thereby resulting in differences in particle alignment. Preeruption topographic relief also would control local flow patterns. Samples of Lava Creek tuff were collected at sites along an arc of about 225° outside the periphery of the caldera, but no relation between remanence direction and geographic location of the collecting site was observed.

Finally, the TRM of the tuff may have been locally influenced by magnetic anomalies associated with underlying rocks. This effect might be important in areas where the Lava Creek tuff was erupted onto or near thick sections of basalt, such as at the Sheepeater Cliffs locality (sites 2YR-94–2YR-97). Nevertheless, most samples in this study were from tuff overlying thick sections of older ash flow tuff or related rhyolite flows, Paleozoic and Mesozoic limestones and sandstones, andesitic mudflows of the Absaroka Volcanic Supergroup, and surficial gravels, lithic types that generally are not associated with strong magnetic anomalies.

The section of Lava Creek tuff at Flagg Ranch (sites 2YR-50 and 2YR-52) is suitable for comparing directions of the two

TABLE 4. Summary of Paleomagnetic Data for Mesa Falls Tuff

| Site | Locality | N | $\bar{H}$ (oe) | $\bar{I}$ | $\bar{D}$ | k | $\alpha_{95}$ |
|---|---|---|---|---|---|---|---|
| 1YR-27 | Mesa Falls | 12 | 100 | -60.8 | 185.9 | 317.1 | 2.4 |
| 2YR-108 | Island Park Reservoir Dam | 7 | 100 | -62.0 | 199.9 | 593.7 | 2.5 |
| 2YR-121 | Route 191 Roadcuts | 6 | 100 | -63.9 | 166.6 | 123.6 | 6.1 |

N Number of specimens
$\bar{H}$ Alternating current demagnetizing field in oersteds yielding least dispersion (highest k value)
$\bar{I}$ Mean inclination, degrees, negative upward
$\bar{D}$ Mean declination, degrees from true north
k Precision parameter (Fisher, 1953)
$\alpha_{95}$ Semi-angle of the cone of 95 percent level confidence

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TABLE 5. Summary of Paleomagnetic Data for Huckleberry Ridge Tuff

| Site | Locality | MMB- | Remarks | N | $\bar{H}$ (oe) | $\bar{I}$ | $\bar{D}$ | k | $\alpha_{95}$ |
|---|---|---|---|---|---|---|---|---|---|
| 2YR-140 | Terrace Mountain | a | VIT | 7 | 100 | -1.2 | 210.9 | 254.6 | 3.8 |
| 1YR-20 | Terrace Mountain | a | DVIT | 3 | 100 | 6.4 | 219.2 | 182.8 | 9.2 |
| 1YR-19 | Terrace Mountain | a | DVIT | 3 | 300 | 1.8 | 220.5 | 84.9 | 13.5 |
| 2YR-42 | Golden Gate | a | DVIT(1.5) | 6 | 50 | 85.3 | 235.1 | 10.2 | 22.0 |
| | | | (5) | 8 | 150 | 42.7 | 255.7 | 3.7 | 33.3 |
| | | | (9.5) | 8 | 150 | 40.1 | 293.2 | 10.2 | 18.2 |
| | | | (14) | 6 | 150 | 75.1 | 255.3 | 19.6 | 15.5 |
| | | | (18) | 4 | 150 | 62.6 | 216.1 | 8.0 | 34.8 |
| | | | (23.5) | 7 | 100 | 53.0 | 228.5 | 11.4 | 18.7 |
| | | | (29.5) | 4 | 50 | 54.0 | 233.9 | 38.7 | 14.9 |
| | | | (37) | 2 | 200 | 11.8 | 212.8 | 84.7 | 27.5 |
| 2YR-43 | Golden Gate | a | DVIT(38.5) | 6 | 200 | 9.8 | 224.1 | 28.9 | 12.7 |
| 2YR-44 | Golden Gate | a | UVP (41) | 6 | 75 | 4.5 | 223.8 | 13.4 | 19.0 |
| | | | (42.5) | 6 | 150 | 13.5 | 225.3 | 41.1 | 10.6 |
| 2YR-45 | Golden Gate | b | DVIT (0.2) | 4 | 150 | -1.6 | 223.6 | 96.3 | 9.4 |
| | | | | | 200 | -4.9 | 218.2 | 81.6 | 7.5 |
| | | | (2.5) | 4 | 75 | 6.0 | 224.1 | 35.5 | 15.7 |
| | | | | | 200 | 10.6 | 222.0 | 49.3 | 13.2 |
| | | | (7.5) | 5 | 300 | 9.8 | 215.6 | 72.8 | 9.0 |
| | | | (14) | 4 | 50 | 15.7 | 219.0 | 46.7 | 13.6 |
| | | | | | 200 | 1.4 | 217.3 | 41.5 | 14.4 |
| | | | (20) | 5 | 100 | -2.0 | 219.5 | 31.4 | 13.9 |
| | | | (24.5) | 5 | 200 | -7.0 | 237.4 | 520.4 | 3.4 |
| | | | (27.5) | 3 | 300 | 13.4 | 222.0 | 68.3 | 15.0 |
| | | | (30.5) | 6 | 75 | -3.0 | 226.4 | 19.1 | 15.7 |
| | | | | | 150 | -1.8 | 235.0 | 21.3 | 14.9 |
| 2YR-46 | Golden Gate | b | DVIT | 12 | NRM | 39.8 | 301.4 | 1.7 | 52.2 |
| 0YR-10 | Mount Everts | a | VIT-SB | 5 | 100 | -1.9 | 214.1 | 178.2 | 5.8 |
| (1YR-16) | | | FA | 5 | 200 | 9.1 | 215.6 | 56.1 | 10.3 |
| | | a | AA | 6 | 50 | -2.4 | 210.9 | 10.9 | 21.2 |
| | | GME | SAND | 1 | 100 | -13.6 | 212.2 | - - | - - |
| | | GME | COB | 4 | NRM | 25.9 | 114.7 | - - | - - |
| 1YR-17 | | a | DVIT | 4 | 200 | -4.8 | 213.3 | 87.2 | 9.9 |
| 1YR-18 | | b | DVIT | 4 | 200 | -24.8 | 239.7 | 15.8 | 23.8 |
| 2YR-54 | Type section | a | VIT (1.0) | 7 | 200 | -3.3 | 212.6 | 170.9 | 4.6 |
| 2YR-55 | Type section | a | DVIT (6) | 6 | 100 | -5.0 | 212.1 | 114.6 | 6.3 |
| | | a | DVIT (15) | 4 | 100 | 4.1 | 213.5 | 55.5 | 12.5 |
| | | a | DVIT (24) | 5 | 100 | -2.7 | 207.0 | 25.5 | 15.5 |
| 2YR-56 | Type section | a | DVIT (28) | 4 | 75 | -4.3 | 214.8 | 4.3 | 50.3 |
| 2YR-57 | Type section | b | DVIT (0.3) | 6 | 100 | -11.8 | 209.2 | 37.4 | 11.1 |
| 2YR-59 | Type section | b | DVIT (7.5) | 7 | 100 | -4.0 | 207.4 | 40.9 | 9.6 |
| | | b | DVIT (15.5) | 7 | 150 | -4.4 | 211.7 | 334.7 | 3.3 |
| 2YR-60 | Type section | c | DVIT | 6 | 100 | -3.4 | 213.7 | 15.8 | 17.4 |
| 2YR-63 | Huckleberry Mountain | c | NPL | 7 | 200 | -6.0 | 211.3 | 12.5 | 17.8 |
| | | c | PL | 3 | 200 | -17.6 | 210.3 | 71.8 | 14.7 |
| 2YR-86 | Huckleberry Mountain | c | FNPL | 5 | 100 | -14.3 | 219.2 | 15.9 | 19.8 |
| | | c | SP | 3 | 200 | -13.8 | 204.8 | 107.6 | 11.9 |
| 2YR-87 | Huckleberry Mountain | c | DVIT | 4 | 200 | -19.2 | 236.8 | - - | - - |
| 2YR-64 | Huckleberry Mountain | b | UVP | 3 | 200 | 3.2 | 221.1 | 156.9 | 9.9 |
| | | | | 5 | 200 | -2.1 | 201.2 | 9.9 | 25.6 |
| 1YR-40 | Lizard Creek | c | DVIT | 4 | 200 | -5.1 | 214.0 | 289.2 | 5.4 |
| 2YR-61 | | c | DVIT | 6 | 200 | -3.6 | 219.1 | 110.1 | 6.4 |
| 2YR-47 | Grayling Creek | a? | DVIT | 6 | 50 | -4.8 | 209.4 | 161.7 | 5.3 |
| 2YR-48 | Grayling Creek | a? | SB | 3 | 400 | -2.3 | 208.6 | 237.4 | 8.0 |
| | | a? | DVIT | 3 | 400 | -2.7 | 202.4 | 664.7 | 4.8 |
| 2YR-104 | Grayling Creek | a? | DVIT | 5 | 200 | 0.1 | 205.3 | 61.2 | 9.9 |
| 2YR-105 | Grayling Creek | a? | DVIT | 3 | 200 | 6.8 | 199.1 | 14.8 | 33.3 |
| 1YR-22 | Grayling Creek | a? | DVIT | 4 | 200 | -10.0 | 217.2 | 209.2 | 6.4 |
| 1YR-23 | Grayling Creek | a | DVIT | 13 | 100 | -7.8 | 226.1 | 24.6 | 8.5 |
| 1YR-24 | Grayling Creek | a | DVIT | 9 | 100 | 46.1 | 221.8 | 12.4 | 15.2 |
| | | | | 5 | 100 | 10.2 | 223.0 | 3.0 | 52.9 |

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TABLE 5. (continued)

| Site | Locality | MMB-Remarks | | N | $\bar{H}$ (oe) | $\bar{I}$ | $\bar{D}$ | k | $\alpha_{95}$ |
|---|---|---|---|---|---|---|---|---|---|
| 1YR-25 | Grayling Creek | b | DVIT | 5 | 100 | -0.1 | 223.5 | 21.4 | 16.9 |
| 1YR-26 | Grayling Creek | b | DVIT | 4 | 100 | 10.7 | 226.8 | 100.7 | 9.2 |
| 2YR-126 | Grayling Creek | ? | DVIT | 3 | 200 | 29.3 | 225.3 | 28.3 | 23.6 |
| 2YR-110 | Ririe Dam | ? | DVIT | 7 | 200 | -5.4 | 215.4 | 127.3 | 5.4 |
| | | | BAS XEN | 4 | 200 | -0.5 | 214.4 | 121.1 | 8.4 |
| 2YR-119 | Warm River | ? | VIT+DVIT | 7 | 200 | -5.3 | 214.2 | 32.7 | 10.7 |
| 2YR-88 | Deadman Bar | c | VP | 3 | 100 | -1.9 | 208.7 | 84.1 | 13.5 |
| 2YR-53 | Moran Quarry | b | DVIT | 6 | 150 | 7.6 | 220.2 | 50.4 | 9.5 |
| 2YR-123 | Lionhead | b | VIT | 6 | 200 | -23.5 | 212.3 | 190.9 | 4.9 |
| 2YR-143 | Badger Creek | ? | DVIT | 6 | 100 | 16.8 | 213.6 | 31.5 | 12.1 |
| 2YR-127 | Casey Lake | ? | DVIT | 6 | 100 | 18.9 | 214.7 | 43.4 | 10.3 |
| 2YR-133 | Big Hole Mountains | ? | DVIT | 3 | 400 | 0.6 | 200.7 | 10.9 | 39.4 |
| 2YR-120 | Ashton Roadcut | ? | DVIT | 8 | 50 | -1.4 | 193.3 | - - | - - |
| 2YR-139 | Crown Butte | a-b | DVIT | 6 | 400 | -7.0 | 210.5 | 117.8 | 6.2 |
| 2YR-138 | Tepee Creek | b | DVIT | 3 | 200 | 27.8 | 233.5 | - - | - - |
| 2YR-136 | Targhee Pass | ? | DVIT | 2 | 400 | 11.4 | 210.7 | - - | - - |
| 137 | Roadcuts | | | | | | | | |
| 2YR-69 70, 71, 72 | Lake of the Woods | ? | DVIT | 7 | 150 | 35.7 | 246.5 | 18.3 | 14.5 |
| 2YR-75 | Reclamation Road | ? | DVIT | 2 | 200 | 38.2 | 240.9 | 997.4 | 7.9 |

MMB indicates member of Huckleberry Ridge Tuff: a, b, and c referring to lower, middle, and upper members, respectively. a? where correlation uncertain. Query where unknown. GME = gravel of Mount Everts.
VIT = vitrophyre
DVIT = devitrified tuff
UVP = upper vapor phase zone
SB = densely welded; blue spherulitic tuff directly above vitrophyre
FA = fused air fall ash
AA = air fall ash
SAND = friable sandstone
COB = cobble of Absaroka Volcanic Supergroup
NPL = non-lineated tuff of upper member
PL = highly lineated tuff of upper member
FNPL = fused non-lineated tuff
SP = stretched pumices
BAS XEN = basalt xenoliths
VP = vapor phase zone
numbers in parentheses are average stratigraphic distance (in meters) above base of exposure (for the lower member at Golden Gate) or above base of member (for all other cases) for groups of specimens.
N number of specimens
$\bar{H}$ alternating current demagnetizing field in oersteds yielding least dispersion (highest k value)
$\bar{I}$ mean inclination, degrees, positive downward
$\bar{D}$ mean declination, degrees from true north
k precision parameter (Fisher, 1953)
$\alpha_{95}$ semi-angle of the cone of 95 percent level confidence

members to gain information concerning the interval of time between eruptions. Mean directions of both members are essentially coincident (Table 3) and thereby do not indicate a major lapse of time separating emplacement of the two subsheets. The time between eruptions cannot be determined from remanence directions alone, but the data are consistent with geologic observation, notably the lack of relief along the contact between the two members, indicating a rapid succession of eruption without intervening time for erosion. Both paleomagnetic and geologic considerations point to the likelihood that eruption of the Lava Creek tuff and development of the Yellowstone caldera took place within a few tens of years.

## Paleomagnetism of the Mesa Falls Tuff

Three site means were determined from a total of 25 specimens, all of which were drilled from the outcrop (Table 4). The mean direction of magnetization for the unit, calculated from remanence directions of four randomly chosen specimens from each of the three sites, is $D = 184.5°$, $I = -64.0°$, and $\alpha_{95} = 4.0$, which defines a VGP at 86.5°S, 132.7°E ($\partial p = 5.0°$, $\partial m = 6.4°$).

Whereas within-site dispersion is extremely small, between-site scatter is marked; the confidence cones at the 95% level for the three sites do not intersect (Table 4). Tectonic tilting is not discernible at any of the three sites, the foliation planes being essentially horizontal. Differences in cooling rates also do not apparently contribute to the between-site dispersion inasmuch as mean directions of rapidly chilled tuff at localities 2YR-108 and 2YR-121 are substantially different.

Sources of between-site dispersion might be partly from local magnetic anomalies associated with underlying rocks and partly from chemical remagnetization. Although samples from sites 2YR-108 and 1YR-27 contain little evidence of chemical alteration, those from site 2YR-121 exhibit replacement of magnetite by hematite, and many of the silicates are stained by hematite. It is possible, then, that CRM has altered the original direction of TRM of samples from site 2YR-121.

## Paleomagnetism of the Huckleberry Ridge Tuff

Inasmuch as initial paleomagnetic study indicated anomalous magnetization within the Huckleberry Ridge tuff, it is paleomagnetically the most thoroughly studied of the Yellowstone Group. The following discussion is based on an analysis

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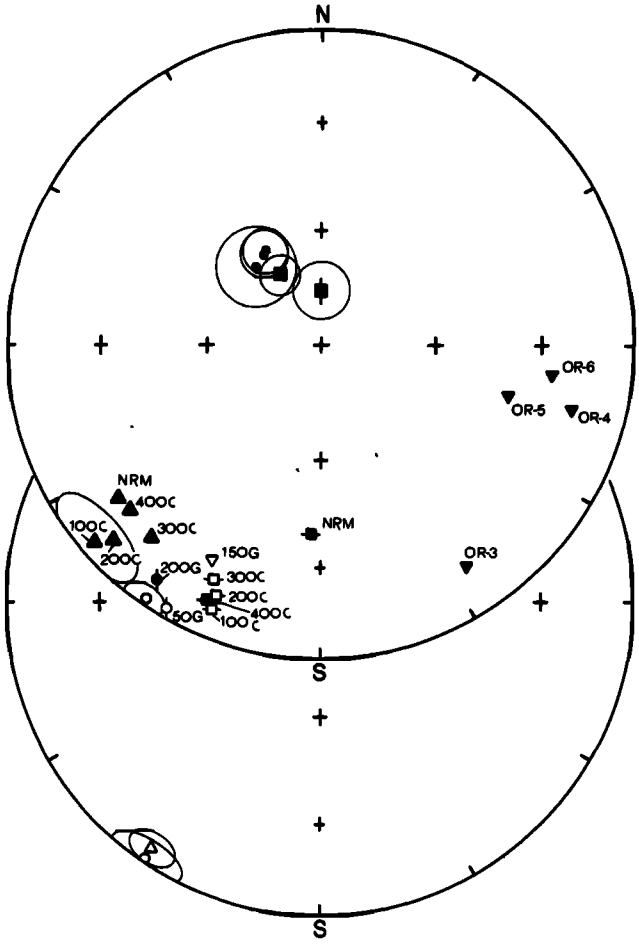



Fig. 14. (Top) Mean directions of remanence of sediments overlain by the Huckleberry Ridge tuff. The open circle is the mean direction of remanence of the Huckleberry Ridge tuff. Solid triangles are the mean directions of remanence of the baked sandstone in the Cody shale at Mount Everts following thermal demagnetization; small squares with horizontal ticks represent mean remanence directions of the Cody shale at Huckleberry Ridge after thermal cleaning. The solid square is the mean remanence direction of the Cretaceous Elkhorn Mountains Volcanics, and the solid square with vertical ticks represents the mean remanence direction of the Cretaceous Boulder batholith [from *Hanna,* 1967, 1973]. Solid circles represent the mean remanence directions of the Cretaceous Niobrara Formation [from *Shive and Frerichs,* 1974]. The solid circle with vertical ticks is the mean direction of fused ash, and the open circle with vertical ticks is the mean direction of air fall ash at Mount Everts (Table 5). Inverted triangles are NRM directions of volcanic cobbles in the gravel of Mount Everts; specimens OR-4, OR-5, and OR-6 are from one hand sample. These directions did not move upon af cleaning. Temperatures in degrees Celsius of thermal demagnetization are denoted by C; values of af demagnetization (in oersteds) are denoted by G. (Bottom) Mean directions after af cleaning of basalt xenoliths (triangle) in the Huckleberry Ridge tuff and of the tuff (circle) at the Ririe Dam locality (sites 2YR-110 and 2YR-111, Table 5).

of 384 specimens from 57 sites representing 23 separate localities (Table 5).

High paleomagnetic dispersion is observed in many of the devitrified portions of this unit. Chemical remagnetization carried by a number of secondary phases is responsible for most of the between-site and within-site dispersion within the Huckleberry Ridge tuff, and the original TRM cannot be confidently isolated by either af or thermal cleaning of many samples of remagnetized tuff. In contrast, the basal vitrophyre, the overlying densely welded spherulitic zone, rapidly chilled margins at partial cooling boundaries within the compound unit, and most tuff at the outer margins of the flow show little dispersion of NRM about a horizontal southwesterly direction. The direction of remanence of these portions of tuff after demagnetization is essentially identical to the NRM direction. Because magnetization directions of devitrified tuff of the flow interior are generally untrustworthy, the unit mean ($D$ = 214.5°, $I$ = −2.2°, and $\alpha_{95}$ = 4.1°) was calculated from three randomly chosen specimens from each of four sites of basal vitrophyre or other rapidly cooled portions of the flow (sites 2YR-140, 2YR-54, 2YR-61, and 2YR-110). The mean direction yields an unusual VGP at 37.2°S, 156.1°W ($\partial p$ = 2.0°, $\partial m$ = 4.1°).

Dissimilar rock types associated with or affected by the Huckleberry Ridge eruption that have magnetic mineralogies different from the mineralogy of the tuff have very similar directions of magnetization, providing evidence that the paleomagnetic mean of the Huckleberry Ridge tuff reflects the paleomagnetic field about 1.92 m.y. ago. Baked Cretaceous sandstone beneath the tuff and basalt xenoliths within the tuff possess very similar horizontal remanence directions (Figure 14). Furthermore, the depositional remanent magnetization of the type B Pearlette ash, found over large areas of western North America, is similar to that of the Huckleberry Ridge tuff [*Reynolds and Larson,* 1972].

Preliminary determinations of the intensity of the ancient geomagnetic field ('paleointensity' measurements) were made on two specimens of Huckleberry Ridge tuff, a basalt xenolith in the Huckleberry Ridge tuff, and two specimens each of the Lava Creek and Mesa Falls tuffs. Paleointensity values were determined by heating specimens in atmospheres with controlled oxygen fugacity according to the procedures of *Wilson* [1961], with the exception of a specimen of vitrophyre from the Lava Creek tuff, which was heated in air following the Thellier-Thellier technique [*Thellier and Thellier,* 1959]. Paleointensity values of specimens from the Huckleberry Ridge tuff, including the basalt xenolith, average 4000 $\gamma$. This contrasts greatly with average paleointensity values of 29,000 $\gamma$ and 32,000 $\gamma$ for the Mesa Falls and Lava Creek tuffs, respectively. These data, together with the anomalous remanence directions of the Huckleberry Ridge tuff, strongly suggest that this ash flow was erupted and cooled during a geomagnetic polarity transition (or major excursion of the field) 1.92 m.y. ago. If it is a transition, it corresponds perhaps to a boundary of the youngest of the Réunion polarity events [see *Grommé and Hay,* 1971].

The mean directions of the three sites in the basal vitrophyre of the lower member are statistically indistinguishable (Figure 15*a*). Mean remanence directions of the middle member also show little variability, with the exception of the vitric tuff of the two-pumice upper part of the middle member in the Henry Mountains, Montana (site 2YR-123). This tuff, representing ash flow material rapidly chilled against steep preeruption topography, possesses an inclination 21° steeper than the mean inclination of the Huckleberry Ridge tuff (Figure 15*b*). The difference in direction may reflect an unknown time difference in cooling, but more likely, it is the result of tectonic tilting or slumping of a large block inasmuch as two-pumice tuffs of the same stratigraphic level elsewhere have means statistically identical to the unit mean of the Huckleberry Ridge tuff. Site means calculated from the three sites in the uppermost member are also very similar, as is shown by intersection of their confidence cones (Figure 15*c*). Declination varies by 10.4°; inclination by only 1.7°.

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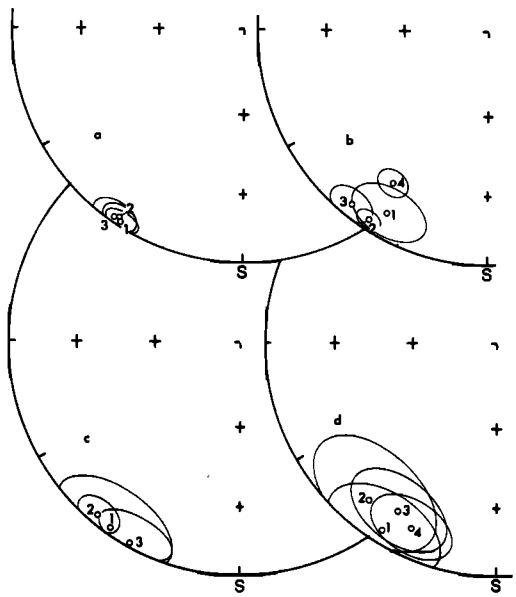


Fig. 15. Mean remanence directions for the Huckleberry Ridge tuff. (*a*) Basal vitrophyre (lower member): 1 represents Terrace Mountain (site 1YR-21); 2, the type section (site 2YR-54); and 3, Mount Everts (site 0YR-16). (*b*) Middle member: 1 represents the base at the type section (site 2YR-57); 2, the top at the type section (site 2YR-59); 3, the base at the Golden Gate (site 2YR-45); and 4, the vitric tuff at the Madison Range (site 2YR-123). (*c*) Upper member: 1 represents the type section (site 2YR-60); 2, Lizard Creek (site 2YR-61); and 3, Deadman Bar, Snake River (site 2YR-88). (*d*) Upper member at Huckleberry Mountain, Wyoming: 1 represents the vapor phase of nonlineated tuff; 2, fused nonlineated tuff; 3, lineated tuff; and 4, stretched pumice fragments.

Mean directions of magnetization of sites in lineated tuff were compared with those in nonlineated tuff within the section of the upper member on Huckleberry Mountain to determine if the lineation, caused by stretched compacted pumice fragments and aligned phenocrysts, was the result of deformation at relatively high temperatures (above the Curie point of magnetite) or at lower temperatures. As is shown in Figure 15*d*, groups of specimens of lineated tuff, stretched and greatly deformed collapsed pumice fragments within the lineated tuff, vapor phase nonlineated tuff, and fused nonlineated tuff all show essentially identical directions of magnetization; within-site dispersion is largely from isothermal remanent magnetization effects. Similar results were obtained following thermal demagnetization of lineated and nonlineated tuff at temperatures up to 550°C. The pronounced lineation therefore has not altered the primary direction of magnetization; apparently, the process that created the extreme lineation occurred at temperatures sufficiently high that the tuff acquired its TRM after deformation.

Mean remanence directions of the basal vitrophyre, the lower and upper chilled margins of the middle member, and the upper member within thick sections of tuff are essentially identical (Figure 16). These comparisons give no indication that large intervals of time relative to PSV separated eruption of the individual members of the Huckleberry Ridge tuff.

Polarity determinations of ash flow tuff at selected localities in the Yellowstone–Island Park–Teton region have fully supported the stratigraphic sequence given by *Christiansen and Blank* [1972]. For example, normal directions appropriate for the Lava Creek tuff were obtained from the vent tuff of *Boyd* [1961] at Mount Everts mapped by Christiansen and Blank as distal end Lava Creek tuff overlying Huckleberry Ridge tuff. Boyd had considered this exposure to be the source of the adjacent welded tuff now recognized as the Huckleberry Ridge tuff.

## SUMMARY

Stable TRM in the three welded tuffs of the Yellowstone Group is primarily carried by very small (<2 $\mu$m) grains of titaniferous magnetite. Nevertheless, the intermediate direction of TRM has been masked in parts of the Huckleberry Ridge tuff by a secondary remanence in a normal direction that is carried by authigenic magnetic minerals. These authigenic ferric oxide phases possess a wide range of coercivities and blocking temperatures; consequently, af and thermal demagnetization of remagnetized samples results in large changes in remanence direction with the removal of the secondary overprint. The direction of TRM cannot be confidently isolated in samples containing secondary magnetic minerals with high coercivities and high blocking temperatures.

Paleomagnetic dispersion recorded in this study is rather similar to that of the Pleistocene Bandelier tuff in New Mexico [*Doell et al.*, 1968], in which confidence cones ($\alpha_{95}$) of site means even within a vertical section do not all intersect. Study of ash flow sheets of the eastern Great Basin also revealed small but significant differences in paleomagnetic directions within flows [*Grommé et al.*, 1972]; the greatest source of dispersion was considered to be uncertainty in determining exact tectonic corrections.

Results from this study do not fully correspond with those of *Nairn et al.* [1975], who concluded that their most reliable paleomagnetic results of welded tuff (least dispersion) came from the devitrified interiors of ash flow sheets. In contrast,

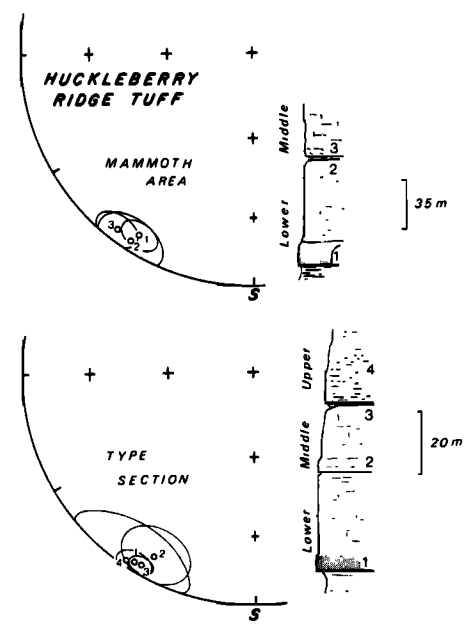



Fig. 16. Mean directions of magnetization of (top) two members of Huckleberry Ridge tuff in the Mammoth area and (bottom) three members at the type section. Mean directions are surrounded by cones of 95% confidence after the af demagnetization level which gave the least dispersion (Table 5). Numbers adjacent to mean directions refer to stratigraphic level in schematic section. Shading denotes vitrophyre and overlying densely welded spherulitic tuff.

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greatest dispersion in the Huckleberry Ridge tuff arises from secondary magnetic phases developed in the initially more permeable devitrified tuff of flow interiors, where groundwater could promote magnetic alteration by hydration, oxidation, movement of iron, and its subsequent deposition as ferric oxide.

These differences indicate that in addition to tectonic tilting and local magnetic anomalies, factors such as permeability of tuff as controlled by degree of welding, availability and chemistry of groundwater and hydrothermal solutions, and intensity of magnetization contribute to variations in the paleomagnetic remanence of ash flow sheets. These factors will vary between and within units on a regional and local scale and may, by the development of CRM with time, contribute to a complex paleomagnetic record.

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